

# SURYAPRAKASH S

+91 7010648069 | suryaprakashstech@gmail.com | **LinkedIn** : <https://www.linkedin.com/in/suryaprakash-s> | **Github** : <https://github.com/suryaprakash-ss>

## EDUCATION

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|---------------------------------------------------------------------------------------|--------------------------|
| <b>KPR Institute of Engineering and Technology</b>                                    | <b>Coimbatore, India</b> |
| B.E., Computer Science and Engineering (Artificial Intelligence and Machine Learning) | 2022 - 2026              |
| <ul style="list-style-type: none"><li>CGPA: 9.06</li></ul>                            |                          |
| <b>Kongu Matriculation Higher Secondary School</b>                                    | <b>Tirupur, India</b>    |
| Higher Secondary Education                                                            | 2020 - 2022              |
| <ul style="list-style-type: none"><li>Percentage: 92.33</li></ul>                     |                          |

## SKILLS

- Programming Languages:** Python, Java, SQL
- Backend Frameworks:** FastAPI, Pydantic
- Cloud & Storage:** AWS S3, boto3, Google Document AI
- PDF & Document Processing:** PyMuPDF, pypdf, pikepdf, ReportLab
- AI & LLM Tools:** Vertex AI, Gemini API, Prompt Engineering, LLM Integration
- Core Concepts:** Data Structures, OOPs, REST APIs, API Integration, Problem Solving

## EXPERIENCE

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| <b>Gen AI Developer Intern - Value Health Inc (Sept 2025 - May 2026)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Coimbatore, Tamil Nadu, India</b> |
| <ul style="list-style-type: none"><li>Developed an AI-powered healthcare automation platform for electronic prior authorization (ePA) workflows.</li><li>Built FastAPI-based REST APIs to process patient, payer, and prescription data for automated authorization systems.</li><li>Implemented PDF AcroForm extraction, annotation rendering, and coordinate normalization using PyMuPDF, pypdf, pikepdf, and ReportLab.</li><li>Developed LLM-based document understanding pipelines using Vertex AI/Gemini for intelligent field-label mapping.</li><li>Built payer-template matching workflows using RapidFuzz and AWS S3 integration for template retrieval and caching.</li><li>Solved production-level PDF engineering issues including rotated pages, annotation mismatches, and overlay alignment.</li></ul> |                                      |

## PROJECTS

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| <b><u>Cross-Lingual Regularization for Enhancing BERT Performance on Low-Resource Languages Without Additional Data</u></b>                                                                                                                                                                                                                                                                                                                                                                                      |  |
| <ul style="list-style-type: none"><li>Proposed a Cross-Lingual Regularization (CLR) approach that aligns multilingual BERT embeddings between high-resource and low-resource languages during fine-tuning.</li><li>Applied consistency loss to bring semantically similar cross-lingual embeddings closer and contrastive loss to separate unrelated sentence representations.</li><li><b>Technologies Used:</b> Python, PyTorch, Transformers, mBERT, FastAPI, Streamlit, Scikit-learn, NumPy, Pandas</li></ul> |  |
| <b><u>Natural Language → SQL Chatbot</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
| <ul style="list-style-type: none"><li>Designed a system that converts natural language questions into executable SQL queries with structured table outputs.</li><li>Enabled multi-step reasoning using LangChain + Groq LLM with ReAct framework and MCP tool integration.</li><li><b>Technologies Used:</b> Python, Streamlit, LangChain, Groq LLM, MCP, PostgreSQL, psycpg2, Pandas</li></ul>                                                                                                                  |  |
| <b><u>Leaf Disease Detection Using Xception</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <ul style="list-style-type: none"><li>Spearheaded the development of a lightweight Xception-based CNN model for real-time agricultural disease detection.</li><li>Achieved 96.5% accuracy and 91.6% precision, outperforming architectures like ResNet50, Inception, and VGG19.</li><li><b>Technologies Used:</b> Python, TensorFlow, OpenCV, Image Processing, Scikit-learn.</li></ul>                                                                                                                          |  |

## PUBLICATIONS

- Hybrid CNN-CatBoost Approach for Predicting Biodegradability of Synthetic Chemicals** - IEEE, 2024
- ECG Arrhythmia Classification Using CNN-BiLSTM Models for Enhanced Diagnostic Accuracy** - IEEE, 2024
- Cross-Lingual Regularization for Enhancing BERT Performance on Low-Resource Languages Without Additional Data** - IEEE 2026

## ADDITIONAL INFORMATION

### ACHIEVEMENTS

- First Place Winners** - Cipher Quest: The Data Mystery at Coimbatore Institute of Technology (CIT), Coimbatore.
- Academic Topper** – Awarded for outstanding academic performance in the 2022-2023 academic year.
- All Rounder of the Department** – Awarded for academic, extracurricular, and leadership excellence on Annual Day 2024.

### CERTIFICATIONS

- Google Crash Course on Python** - Coursera
- IBM Machine Learning Specialization** - Coursera

### LEADERSHIP & PROFESSIONAL MEMBERSHIPS

- Secretary** – NeuraNest – Students Association, KPR Institute of Engineering and Technology.
- IEEE Executive Member** – IEEE CIS Society, KPR Institute of Engineering and Technology.